

Physics 5C Discussion (Worksheet 10)

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Click for the [Zoom link](#) and [solutions to the worksheets](#).

Problem 1: The oscillation of the electric field and the magnetic field together “generates” a propagating electromagnetic wave. Take the speed of light as $c = 3 \cdot 10^8$ m/s.



A “ $\perp$ -mobile” antenna is emitting cell-phone signals. The antenna is oriented vertically, and it emits a signal that travels horizontally outward from the antenna. Jinyan is standing due east of the antenna, and at his location, the signal is traveling eastward. At one instant, Jinyan measures the electric field at his location and finds that the field is pointing straight up with magnitude 3 V/m.

Part (a) What are the direction (west? east? south? north? up? down?) and magnitude (numerical) of the magnetic field at Jinyan’s location at that instant?

Part (b) The signal that Jinyan receives has a wavelength $\lambda = 1$ m. If Jinyan’s iPhone-13 is receiving a very steady signal throughout the time, find the direction and magnitude of electric field, and that of magnetic field at his location $\lambda/(2c)$ seconds after.^{1 2}

Part (c) In this part, treat the antenna as a point source emitting wave signal in all directions. If Jinyan is standing 1 km from the antenna and 3 V/m is the amplitude of the oscillating electric field at his location, estimate the intensity of the cell-phone signal emitted by the antenna at the location 1 m from the antenna.

¹This is a challenging problem, but it turns out that you don’t need to do any numerical computation here. You just need to figure out what I am asking, and recall some knowledge about waves.

²Using $c = \lambda f$ and $f = 1/T$, find T in terms of c and λ . Then determine what $\lambda/(2c)$ means.

The measured electric field at Jinyan's location is polarized in the vertical direction.

Part (d) Jinyan then uses two polarizing sheets to change the polarization of the wave signal. The first one is oriented $\theta_1 = 30^\circ$ from the vertical, and the second one is oriented $\theta_2 = 90^\circ$ from the vertical. The two sheets are put one after another along the line of the wave transmission such that the wave signal will pass through the first one and then the second one. Find the intensity of the wave after it has passed through the second polarizer.

Part (e) Jinyan's evil friend Michael sneakily switched the order of the two polarizers in part (d). Now what intensity of the wave will Jinyan measure after the wave has passed through the second polarizer?